

Algorithmic Pseudocode Sample 17

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Input: Set of sequences(S)
Output: Distance Matrix(D)

```
1: for  $s_1$  in S do
2:    $Es_1 \leftarrow \text{encoded } s_1$ 
3:    $Cs_1 \leftarrow \text{Gzip compressed } Es_1$ 
4:    $Ls_1 \leftarrow \text{length of } Cs_1$ 
5:    $D\_local \leftarrow [ ]$ 
6:   for  $s_2$  in S do
7:      $Es_2 \leftarrow \text{encoded } s_2$ 
8:      $Cs_2 \leftarrow \text{Gzip compressed } Es_2$ 
9:      $Ls_2 \leftarrow \text{length of } Cs_2$ 
10:     $s_1s_2 \leftarrow \text{Concatenate}(s_1, s_2)$ 
11:     $Es_1s_2 \leftarrow \text{encoded } s_1s_2$ 
12:     $Cs_1s_2 \leftarrow \text{Gzip compressed } Es_1s_2$ 
13:     $Ls_1s_2 \leftarrow \text{length of } Cs_1s_2$ 
14:     $NCD \leftarrow \frac{Ls_1s_2 - \text{Min}(Ls_1, Ls_2)}{\text{Max}(Ls_1, Ls_2)}$ 
15:     $D\_local.append(NCD)$ 
16:   end for
17:    $D.append(D\_local)$ 
18: end for
19: return D
```
